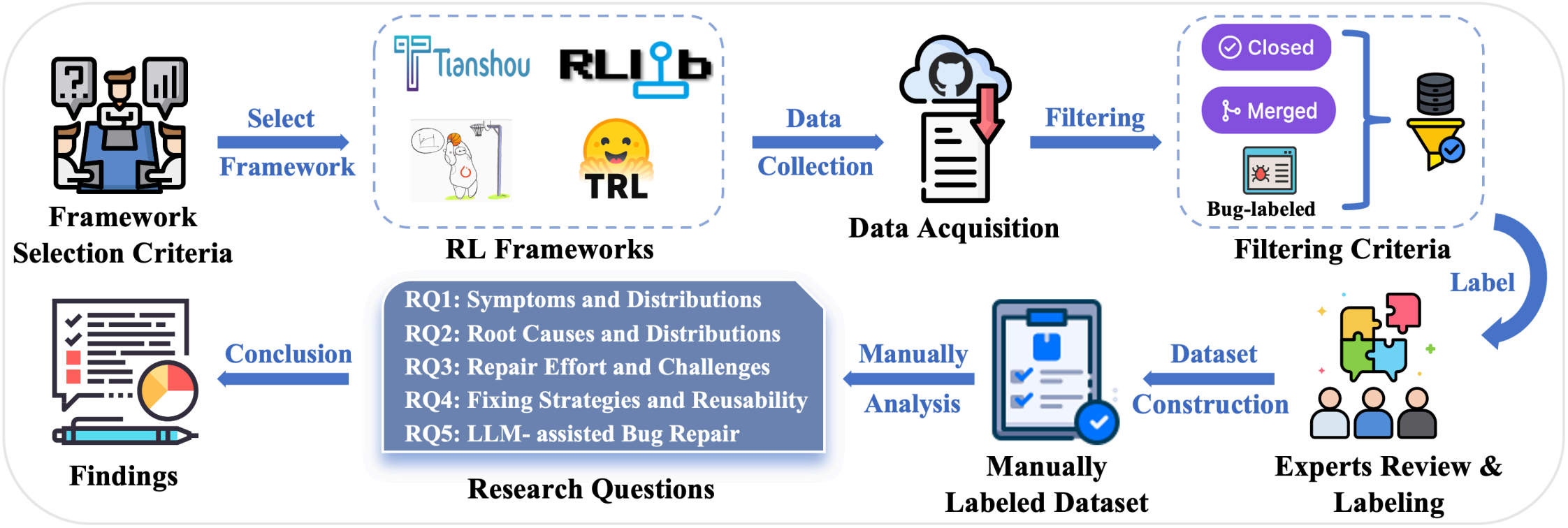

Framework Selection Criteria
Select Framework
Tianshou
RLlib
TRL
RL Frameworks
Data Collection
Data Acquisition
Filtering
Closed
Merged
Bug-labeled
Filtering Criteria
Label
Experts Review & Labeling
Dataset Construction
Manually Labeled Dataset
Manually Analysis
RQ1: Symptoms and Distributions
RQ2: Root Causes and Distributions
RQ3: Repair Effort and Challenges
RQ4: Fixing Strategies and Reusability
RQ5: LLM- assisted Bug Repair
Research Questions
Conclusion
Findings